SALAR ABBASPOURAZAD

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WORK EXPERIENCE

Apple Inc., Machine Learning Scientist *Sep 2022 - Present*

- Lead pre-training, evaluation and internal distribution of multi-modal foundation models for health and wearable data (example). Scaled the project from pure research to production, improving accuracy of existing Apple health features, enabling new production features, and shipped on Apple devices as of Fall 2025
- ★ Owner and lead developer of a general-purpose infrastructure for data processing, distributed training, and evaluation (used internally by >10 research/production teams)
  - Scalable dataset preprocessing/sharding and fast/memory-efficient dataloading
  - Config-driven infrastructure for various pre-training frameworks (contrastive learning, joint-embedding architectures, auto-regressive, masked autoencoding, knowledge distillation, etc), fine-tuning methods (e.g., LoRA), architectures (Transformer-based, CNN-based, etc), multi-node training/inference on Apple secure cloud for health data
  - While being general-purpose, most training methods and architectures are tailored for sequential and time-series data, and all are written from scratch in native `pytorch` & `pyspark`
- Develop health intelligence and insight generation via agents and multi-modal post-training of large language models (LLM s) for time-series wearable data
- Published 7 papers including ICLR , ICML and Nature Comm . Filed 2 patents. Co-mentored 6 interns

Apple Inc., Machine Learning Intern *May 2021-Aug 2021*

Developed contrastive self-supervised learning methods for signals recorded from wearable devices

EDUCATION

PhD, Electrical & Computer Engineering, University of Southern California, Los Angeles *2015 - 2022*
Thesis: Dynamical representation learning for multiscale brain activity
Developed deep learning and statistical machine learning models for brain machine interfaces
Published 3 papers in Nature journals

MSc, Computer Science, University of Southern California, Los Angeles *2018 - 2020*
GPA: 3.97/4.0
Track: Artificial Intelligence

BSc, Electrical Engineering, Sharif University of Technology, Tehran, Iran *2010 - 2015*
GPA: 18.92/20 (equivalent to 4.0/4.0, class of 2015 top 3 students)
Minor in Economics

HONORS AND ACHIEVEMENTS

Ranked **2nd** among more than 300,000 competitors in the nationwide university entrance exam , Iran *2010*

RESEARCH INTERESTS

Deep representation learning, Foundation models pre-training & post-training, ML for time series

SELECT PUBLICATIONS (GOOGLE SCHOLAR)

E. Erturk*, F. Kamran*, **S. Abbaspourazad**, et al., "Beyond Sensor Data: Foundation Models of Behavioral Data from Wearables Improve Health Predictions", ***ICML***'25 (link)
A. Miller*, J. Futoma*, **S. Abbaspourazad**, et al., "A wearable based aging clock associates with disease and behavior", ***Nature Comm.***, 2025 (link)
S. Abbaspourazad, et al., "Wearable Accelerometer Foundation Models for Health via Knowledge Distillation", *arXiv*, 2024 (link)

S. Abbaspourazad et al., "Large-scale training of foundation models for wearable biosignals", ***ICLR***'24 (link)
S. Abbaspourazad*, E. Erturk*, et al., "Dynamical flexible inference of nonlinear latent structures in neural population activity", ***Nature Bio. Eng.***, 2023 (link)(highlighted on the cover, see nature coverage)
S. Abbaspourazad, et al., "Multiscale low-dimensional motor cortical state dynamics predict naturalistic reach-and-grasp behavior", ***Nature Comm.***, 2021 (link)(USC story)
O. G. Sani, **S. Abbaspourazad**, Y. Wong, B. Pesaran and M. Shanechi, "Modeling behaviorally relevant neural dynamics enabled by preferential subspace identification", ***Nature Neuro.***, 2020 (link)(USC story)

TECHNICAL STRENGTHS

Modeling	Deep Learning, Self Supervised Learning, Time-series and Sequential Data
Tools	Python, PyTorch, Tensorflow, Spark, Airflow, SQL, C++

SIDE PROJECTS (WITH CLAUDE CODE)

beobeo	Personal portfolio tracker built via a fully automated self-improving loop: user agents test the app and file feedback, developer agents implement fixes
sonarch.ai	Engineered website's agentic RAG system

SERVICE

Co-chair of Machine Learning Summit at Apple
Reviewer for Nature, ICLR, ICML, Neurips, IEEE

INTERESTS

Soccer (former USC & Sharif Uni. Futsal team member), Traveling, Books, Games